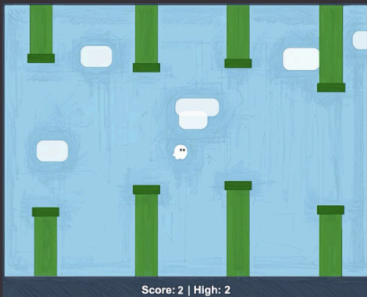


Initialize a local Git repository

```
jane@James-MacBook-Air Lab13-Kiro % curl 'https://static.us-east-1.prod.workshops.aws/public/bc958638-76d3-42b6-8318-06163ddcb578/assets/kiro-introduction-starter-kit.zip' -o kiro-introduction-starter-kit.zip
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100  987k  100  987k    0     0  2578k      0 --:--:-- --:--:-- --:--:-- 2585k
jane@James-MacBook-Air Lab13-Kiro % unzip -o kiro-introduction-starter-kit.zip -d . && rm kiro-introduction-starter-kit.zip
Archive:  kiro-introduction-starter-kit.zip
  creating: ./assets
  inflating: ./assets/ghosty.png
  inflating: ./assets/jump.wav
  inflating: ./assets/game_over.wav
  creating: ./img
  inflating: ./img/example-ui.png
  inflating: ./LICENCE.md
  inflating: ./README.md
  inflating: ./gitignore
jane@James-MacBook-Air Lab13-Kiro % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to 'main' in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/jane/Desktop/Lab13-Kiro/.git/
jane@James-MacBook-Air Lab13-Kiro % git add .
git commit -m 'Initial commit'
[master (root-commit) dced994] Initial commit
 7 files changed, 83 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 LICENCE.md
 create mode 100644 README.md
 create mode 100644 assets/game_over.wav
 create mode 100644 assets/ghosty.png
 create mode 100644 assets/jump.wav
 create mode 100644 img/example-ui.png
```

Create application specifications

I would like to create `Flappy Kiro`, a retro game that will run in a browser. The game will be a endless scroller, where you guide a ghost character through a set of pipes. It should look something like the attached screenshot.



Kiro

The screenshot looks great — a ghost navigating through green pipes with a score bar at the bottom. This is clearly a new feature, but let me confirm before we dive in.

Build a Feature

Requirements

Refine the requirements

Clouds should be semi transparent, and move at different speeds to give the appearance of perspective.

 **Kiro**

 Including Steering Documents

 `interview-coach.md`

 Accepted edits to  `requirements.md`

Updated. Here's what changed in the cloud requirements:

- Each cloud is now assigned a random scroll speed between **0.8 and 2.5 px/frame** when spawned, giving the parallax depth effect
- Clouds are rendered with **semi-transparent fill at 0.4–0.8 opacity** so the background shows through

Everything else stays the same. Happy with this, or anything else to tweak before we move on to the design?

(Optional) Example: Continue to refine the specification

Add comprehensive physics system including: Ghosty's gravity constant, ascent velocity on spacebar press, terminal velocity limits, momentum conservation, and smooth movement interpolation. Include detailed obstacle generation: wall pair spacing, gap size, random gap positioning, wall movement speed, and progressive speed increase. Enhance collision detection with: precise hitbox definitions, wall collision boundaries, ground and ceiling detection, collision response animations, and invincibility frames. Expand game state management: main menu with high score display, gameplay state with real-time scoring, pause functionality, game over screen with restart option, and persistent score storage. Add audio and visual feedback: sound effects for flapping, scoring, collisions, background music, screen shake on collision, particle trails for Ghosty, and visual score indicators.

 **Kiro**

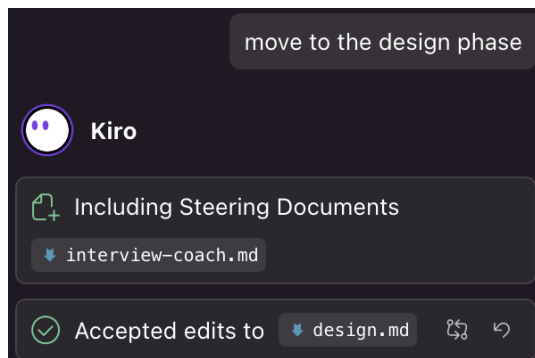
Good set of additions. Let me get those worked into the requirements.

 Including Steering Documents

1. Make a commit of requirements

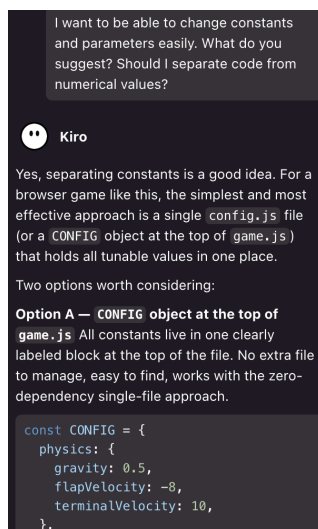
```
[git commit -m 'Produced requirements'  
[master ff8fb81] Produced requirements  
2 files changed, 210 insertions(+)  
create mode 100644 .kiro/specs/flappy-kiro/.config.kiro  
create mode 100644 .kiro/specs/flappy-kiro/requirements.md
```

2. Tell Kiro to move to the design phase in the chat panel.



Refine the design

1. Ask Kiro about where to store code parameters and constants.



(Optional) Add additional design features

Add precise detection algorithms: circular collision for Ghosty, rectangular bounds for walls, ground/ceiling detection. Use optimization guidelines including a target of 60 FPS, efficient sprite batching and memory management for obstacle pooling.



Kiro

Good additions. One thing worth flagging before updating the design: the requirements currently specify AABB (rectangular) collision for Ghosty. Switching to circular collision is a design-level decision that would make the hitbox slightly more forgiving and accurate to the round ghost shape, but it's a deviation from what's written.

My suggestion: use a **circle for Ghosty vs pipes** (radius ~14px, centered on the sprite) and keep **AABB for pipe bounds** — this is

Complete the design

```
jane@Janes-MacBook-Air Lab13-Kiro % git add .
[git commit -m 'Produced design'
[master 7e3dd43] Produced design
 2 files changed, 662 insertions(+), 10 deletions(-)
 create mode 100644 .kiro/specs/flappy-kiro/design.md
```

Refine the implementation tasks

Generate `game-config.json` with complete physics parameters: gravity (800 px/s²), jump velocity (-300 px/s), wall speed (120 px/s), gap size (140px), wall spacing (350px). Create `ghosty-sprites.md` with detailed character specifications: sprite dimensions (32x32px), animation frames for idle/flap/death states, hitbox radius (12px). Develop `audio-assets.md` with sound design specifications: flap sound (short whoosh, 0.1s), score sound (pleasant chime, 0.2s), collision sound (soft thud, 0.3s). Generate `ui-mockups.md` with interface designs: main menu layout with play/high scores buttons, in-game HUD with current score display, game over screen with final score and restart option.



Kiro

Before creating those files, a few things worth flagging:

Physics unit mismatch — the current design uses frame-based physics (gravity 0.5 px/frame², jump -8 px/frame at 60fps). The values you've listed use seconds (800 px/s², -300 px/s). These are equivalent if converted correctly ($800/3600 \approx 0.22$ px/frame² —

Complete the tasks

Generate a commit message and commit to Git. Commit the changes to Git. Look at the changes and create a git commit message to show what's changed. Highlight it's the end of the implementation phase.



Kiro

Let me check what's changed first.

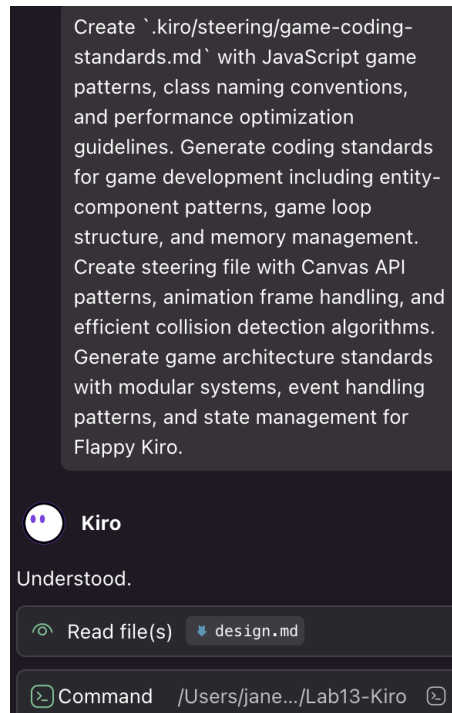
```
Command /Users/jane.../Lab13-Kiro
```

```
git -C . status && git -C . diff --st
```

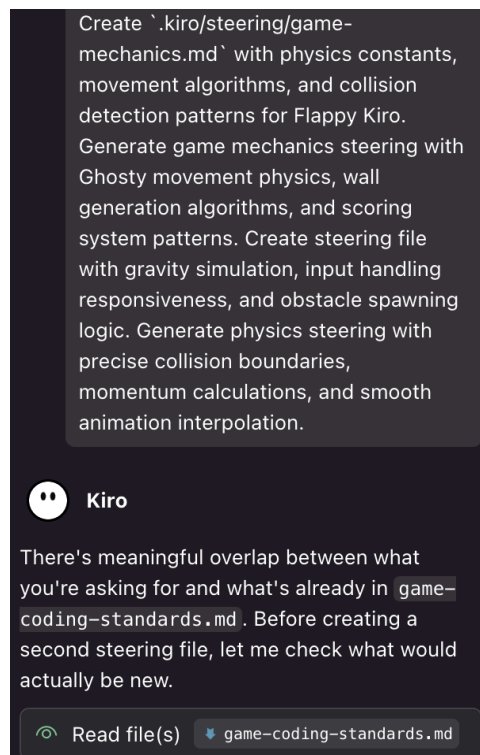
```
On branch master
```

```
Changes not staged for commit:
```

Create game coding standards



Create game mechanics guidelines



Create visual design guidelines

Create ``.kiro/steering/visual-design.md`` with sprite rendering patterns, animation systems, and particle effect guidelines. Generate visual design steering with Ghosty character animations, wall textures, and background parallax effects. Create steering file with Canvas drawing optimization, sprite atlas usage, and visual feedback patterns. Generate audio-visual steering with sound effect integration, screen shake mechanics, and UI animation patterns.



Kiro

Same situation as the last request — a lot of what you've listed (Canvas drawing, screen shake, sprite rendering, particle effects, animation) is already in `game-coding-standards.md`. Creating a duplicate would bloat every AI context window with redundant rules.

Here's what's genuinely not covered yet and would add real value in a visual design steering file:

- **Colour palette** — exact hex values for the

Create game domain knowledge

Create ``.kiro/steering/flappy-kiro-domain.md`` with game state management, score persistence, and difficulty progression patterns. Generate Flappy Kiro domain steering with obstacle generation rules, gap sizing algorithms, and progressive speed increases. Create steering file with game session management, high score tracking, and restart functionality patterns. Generate domain steering with Ghosty behavior patterns, wall collision responses, and game over state handling.



Kiro

Same pattern here — game state management, collision responses, game over handling, and difficulty progression are all already documented between `game-coding-standards.md` and the spec files.

The genuinely uncovered ground across all three recent requests (game-mechanics, visual-design, and this one) is actually the same short list of things. Rather than three separate files that each partially overlap with existing content, one focused file would cover all the real gaps cleanly.

Test steering influence

Create a Ghosty character class with physics properties, animation states, and collision detection following the game standards. Generate a `WallObstacle` class with positioning, movement, and collision boundaries using the coding guidelines. Create a `GameEngine` class with main loop, state management, and rendering system following our architecture patterns. Generate a `ScoreManager` component with point tracking, high score persistence, and display formatting using our domain knowledge.



Kiro

This is implementation work — let me run it through the tasks properly rather than generating code ad-hoc. That way the task statuses stay accurate and the implementation follows the spec.



Read file(s)



tasks.md

Run the game

Start Flappy Kiro and check the application is running.



Kiro

The game is a static HTML/JS file with no server needed. Let me verify the files are in order first, then open it.

 Command /Users/jane.../Lab13-Kiro 

```
ls -la index.html game.js assets/
```

```
-rw-r--r--@ 1 jane  staff  22743
```

```
Apr 18 02:00 game.js
```

```
-rw-r--r--@ 1 jane  staff    739
```

```
Apr 18 01:55 index.html
```

```
assets/:
```

```
total 976
```

```
drwxr-xr-x  5 jane  staff    160
```

```
Nov 20 08:28 .
```

```
drwxr-xr-x 12 jane  staff    384
```

```
Apr 18 01:56 ..
```

 Command /Users/jane.../Lab13-Kiro 

```
node --input-type=module << 'EOF'
```

